

Doubles for both triple reconstruction and assay validation

torchcell lab notes

Script `experiments/010-kuzmin-tmi/scripts/construction_validation_doubles.py`
Data `experiments/010-kuzmin-tmi/results/construction_validation_doubles.csv`
Dendron `[[experiments.010-kuzmin-tmi.scripts.construction_validation_doubles]]`

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1 Why the pure set-cover was the wrong objective

The essential wet-lab task is a small set of double mutants that both (a) reconstructs many top-ranked triples and (b) has **variance in DMF / interaction** to validate the new echo-plating assay against (versus plate-stamping). The triple-coverage set-cover (`[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`) does (a) with 8 doubles but fails (b): those 8 are all **near-neutral** — Costanzo DMF span 0.15, $|\varepsilon| \leq 0.039$, and **zero significant interactions** — so the assay would have almost nothing to discriminate. An assay validated only on near-1.0 fitness and null interactions has not been shown to detect signal.

2 Two tiers, unioned (13 doubles)

coverage (8) the greedy set-cover doubles; reconstruct **all 31** within-10 top- k triples.

validation (5) added for dynamic range and real signal: the doubles with a **significant Costanzo interaction** ($P < 0.05$ and $|\varepsilon| > 0.08$, both signs), the **highest-DMF** double, and a **low-DMF anchor chosen for quality** — among low-DMF (< 0.75) doubles, the one that also reconstructs triples with the tightest SD (YER079W+YJR060W: DMF 0.610 ± 0.018 , covers 3 triples), *not* the noisy DMF-minimum (YGL087C+YJR060W was 0.513 ± 0.172). *novel* (1, *separate*) **YPL046C+YPL081W (ELC1+RPS9A)**, the one within-10 pair with **no DMF in Costanzo or Kuzmin** — a construction candidate to fill the last cell (Section 6).

Result: 31/31 triples still reconstructed, DMF range **0.610–1.178**, ε from **−0.130** to **+0.098**, **3 significant interactions**.

#	Double	Tier	DMF $\pm$ SD	ε	p	sig	triples
1	YER079W+YJR060W	validation	0.610 ± 0.018	−0.003	0.454		3
2	YER079W+YPL081W	validation	0.863 ± 0.098	−0.130	0.035	✓	0
3	YJR060W+YKL033W-A	validation	0.871 ± 0.023	−0.082	0.007	✓	0
4	YJR060W+YPL046C	coverage	0.966 ± 0.018	+0.003	0.453		4
5	YDR057W+YLL012W	coverage	1.000 ± 0.069	−0.036	0.296		5
6	YLR312C-B+YPL081W	coverage	1.016 ± 0.042	−0.019	0.355		3
7	YLL012W+YPL046C	coverage	1.025 ± 0.053	−0.010	0.429		5
8	YDR057W+YPL081W	coverage	1.035 ± 0.045	+0.039	0.260		3
9	YDR057W+YER079W	coverage	1.096 ± 0.054	+0.012	0.435		5
10	YER079W+YLR312C-B	coverage	1.099 ± 0.039	−0.027	0.290		5
11	YBR203W+YPL046C	coverage	1.118 ± 0.017	+0.036	0.077		5
12	YDR057W+YGL087C	validation	1.137 ± 0.024	+0.098	0.001	✓	3
13	YDR057W+YLR312C-B	validation	1.178 ± 0.026	+0.047	0.119		1
14	YPL046C+YPL081W	novel	—	—	—		0

Fig. 1 The 13 selected doubles by tier, with Costanzo DMF, SD, ε and p .

3 Single mutants to inoculate (10)

To build the doubles above, cross these 10 single-mutant strains pairwise. **Every** panel-10 gene appears in at least one double, so all 10 are required — no smaller set builds them all. The # doubles column is how many each gene crosses into (YOS9/YDR057W is the busiest hub at 5; COS111, MMS2 and YKL033W-A each appear once). The novel pair adds no new strain (ELC1 and RPS9A are already in the set).

#	Systematic	Common	# doubles
1	YBR203W	COS111	1
2	YDR057W	YOS9	5
3	YER079W	—	4
4	YGL087C	MMS2	1
5	YJR060W	CBF1	3
6	YKL033W-A	—	1
7	YLL012W	YEH1	2
8	YLR312C-B	(SPH1 locus)	3
9	YPL046C	ELC1	4
10	YPL081W	RPS9A	4

Fig. 2 The 10 single-mutant strains to inoculate, with the number of selected doubles each participates in.

All 44 measured within-10 doubles ranked by DMF, coloured by tier (coverage, validation, unused; significant ε ringed) — the tiered companion to `constructed_10_dmf_forest`, which predates the validation tier and marks only the 8 set-cover doubles. The 45th pair, YPL046C+YPL081W, is unmeasured and so absent here (Section 6).

DMF, SD, SE, ε and p are the published Costanzo values (SD = sample SD over 4 colonies, directly comparable to the assay’s colony SD — see `[[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]`). The low-DMF end is dominated by CBF1/YJR060W (its single defect), so a low DMF there is a fitness effect, not an interaction.

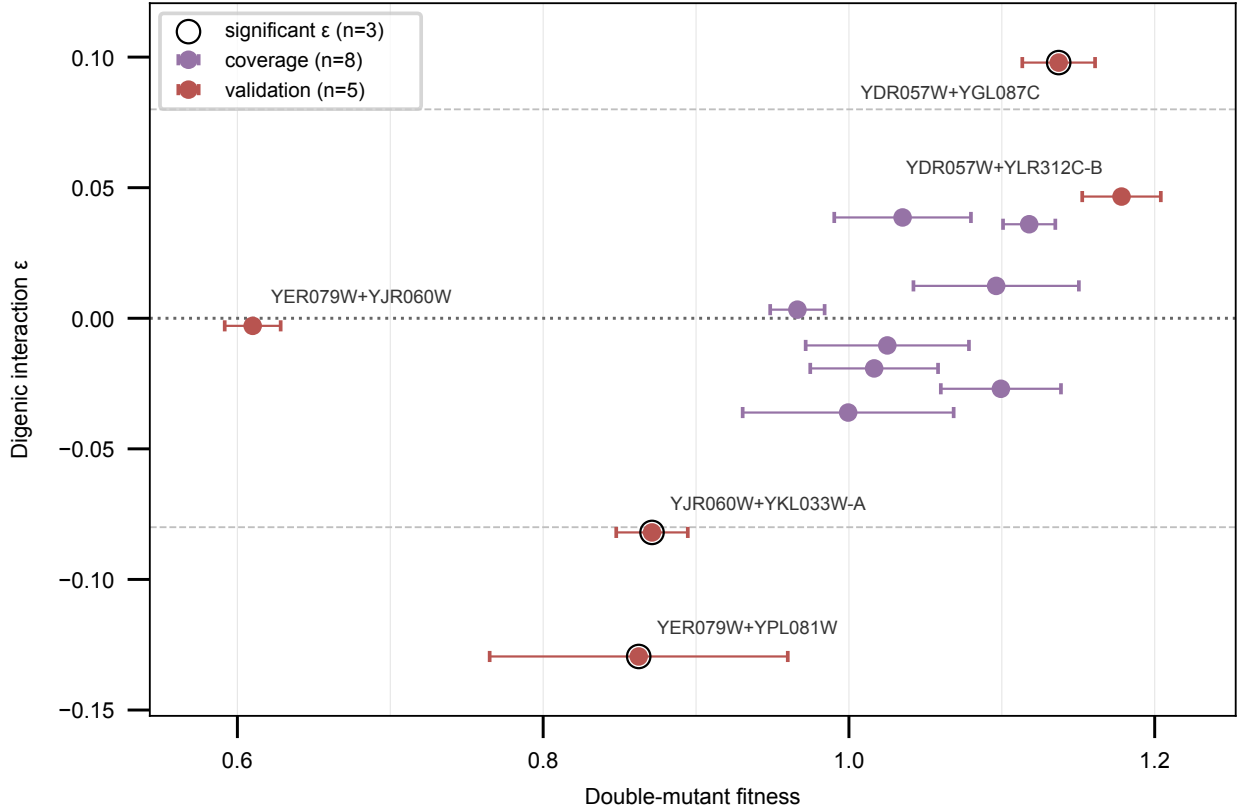


Fig. 3 The selected doubles as a gene-by-gene matrix.

Two of the three significant doubles (YER079W+YPL081W, YJR060W+YKL033W-A) enable no top- k triple — they are pure validation targets; the low anchor YER079W+YJR060W and the strongest-positive- ε YDR057W+YGL087C each also rebuild triples. The set is tunable: drop the two triple-less validation doubles for an 11-double set, or add more near-significant ε .

4 There are only 3 significant interactions in the whole panel

Surveyed across all 66 panel-12 doubles and all three sources: exactly **3** clear Costanzo’s bar ($P < 0.05$ and $|\varepsilon| > 0.08$), and all 3 are within-10 and already in the validation tier — YDR057W+YGL087C (+0.098), YJR060W+YKL033W-A (−0.082), YER079W+YPL081W (−0.130). None hide in the dropped-gene doubles; Kuzmin 2018/2020 have too few measured pairs to reach significance. So “construct the significant ones” means these 3, already selected. The panel is near-neutral by design, so strong interactions are scarce.

4.1 Why a big interaction can still be “not significant” — and why that is fixable

A double’s ε is a point estimate; whether it counts as significant depends on how big it is *relative to its error bar*. The error bar on ε is driven by the colony-to-colony scatter of the double *and* of the two singles it is compared against (all $n = 4$ colonies in Costanzo). So a double sitting on a noisy strain (CBF1, RPS9A) can show a sizable ε that is still inside its own error bar — insignificant. The reverse also happens: YBR203W+YDR057W has a tiny ε (+0.062) that *is* significant ($p = 1.4 \times 10^{-5}$) only because its colonies were very tight (SD 0.010).

The key point for us: **that error bar is set by how many colonies we plate, and that is our choice**. More colonies gives a smaller error bar, shrinking with the square root of the count. So a “noisy but interesting” double is not disqualified — it just needs to be plated deeper. Our goal is depth (measure a few predicted doubles precisely), not breadth (screen many strains shallowly), which is exactly what echo dispensing is good at. So we pick the validation doubles for *interesting values* (fitness extremes, large ε) and drive each to a tight measurement at the bench.

One caveat: deeper plating tightens *our* error bar, but Costanzo’s stays at $n = 4$. Once ours is tighter, a disagreement may be the reference’s noise, not ours.

All 44 measured within-10 doubles by DMF
color = tier; ring = significant ϵ

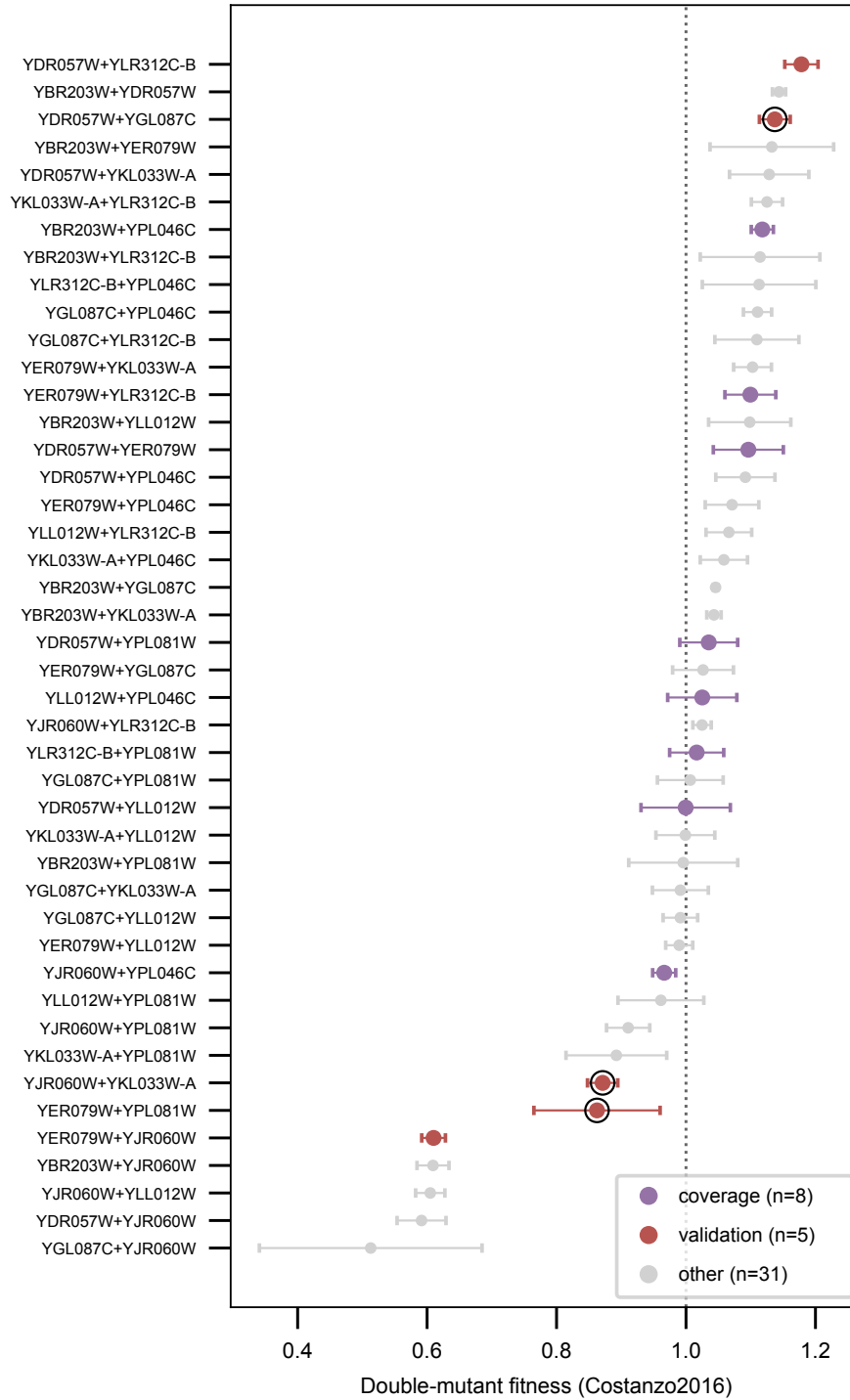


Fig. 4 Forest plot of all 44 measured within-10 doubles, ranked by Costanzo DMF and coloured by tier.

5 Cross-dataset check — Costanzo versus Kuzmin DMF (a real caveat)

The DMF values above are **Costanzo only**. Where Kuzmin also measured a within-10 double, the two references sometimes **disagree sharply**, especially for the low-DMF CBF1 doubles:

double	Costanzo DMF	Kuzmin DMF
YJR060W+YLL012W	0.605	1.119 (K2020)
YJR060W+YPL046C	0.966	0.738 (K2018)
YBR203W+YJR060W	0.609	0.768 (K2018)
YGL087C+YPL046C	1.111	0.998 (K2018)

CBF1/YJR060W’s low DMF is largely a *Costanzo* phenomenon — Kuzmin reads those pairs higher, and YJR060W+YLL012W disagrees by about 0.5. So the low-DMF “signal” our assay would validate against is reference-dependent; treat the low-DMF anchor’s target loosely and, where a Kuzmin value exists, report both. The full per-double Costanzo+Kuzmin comparison is in the supplementary table (Section 7).

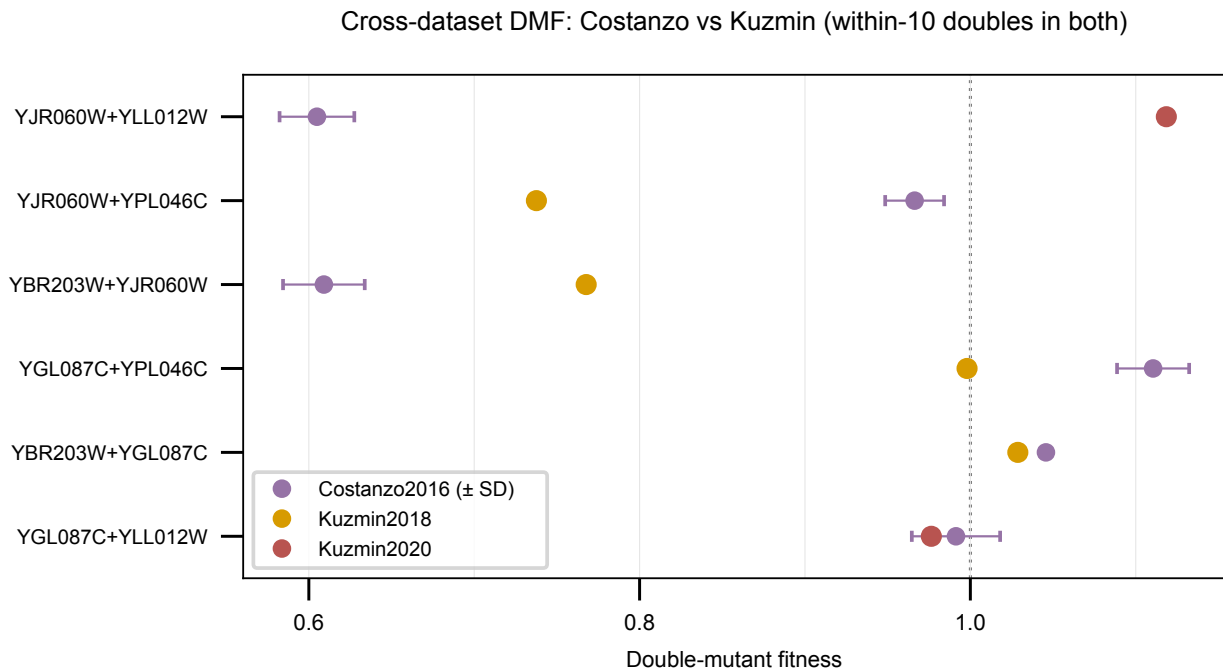


Fig. 5 Dumbbell of every within-10 double measured in both references: Costanzo with SD, joined to the Kuzmin value.

5.1 The same ambiguity exists *within* Costanzo, at the strain level

The disagreement above is between *datasets*. There is a second, easier-to-miss ambiguity *inside* Costanzo: a gene can carry more than one strain, with materially different measurements. YJR060W (CBF1) is the case that matters for this panel —

Costanzo strain	SMF	stddev
YJR060W_dma2646	0.9230	0.0102
YJR060W_sn154	0.5900	0.1138

— two published single-mutant values 0.33 fitness units apart. The queried reference table the assay compares against picked **sn154** by a dictionary collision in `12_panel_inference_3_queried_data_tables.py` (the gene-keyed index is overwritten by each successive record, so the surviving row is whichever LMDb iteration reached last), not by any stated rule. The same code applies no temperature filter, so 26 °C versus 30 °C is also last-record-wins — moot for this panel, where all 12 ORFs happen to carry identical 26/30 entries, but not moot in general (17% of Costanzo strains differ between the two temperatures).

This matters because CBF1 anchors the low end of every fitness correlation we report. It needs a deliberate strain-selection rule before those correlations carry weight.

6 Novel construction candidate — the one unmeasured pair

Of $\binom{10}{2} = 45$ pairs, **44 have a Costanzo DMF; 1 does not: YPL046C+YPL081W (ELC1+RPS9A)** — and it is also absent from Kuzmin 2018/2020. Missing is not the same as synthetic-lethal: an SL pair would show as a measured low fitness or an explicit flag, not a blank. A NaN just means the pair was never in a scored query×array orientation.

SL check (SynthLethDB Yeast_SL.csv). **not synthetic-lethal.** ELC1/YPL046C appears in **zero** SL records, and RPS9A/YPL081W is SL only with its paralog **RPS9B** — the classic redundant-duplicate case — not with ELC1.

BioGRID / SGD second source. Checked via SGD’s aggregated interaction API: ELC1 has **97** recorded genetic-interaction partners and **RPS9A is not among them** — no interaction of any type (SL, negative or positive genetic, synthetic growth defect). ELC1 is well studied at 97 partners, so RPS9A’s absence is a real gap, not sparse coverage.

Recommendation. Construct it. It is the one cell of the 10×10 matrix with no value in Costanzo, Kuzmin, SynthLethDB or BioGRID, and it is not synthetic-lethal, so it should be buildable. Tagged **novel** in the supplementary table.

7 Supplementary table — all 45 within-10 doubles

`experiments/010-kuzmin-tmi/results/construction_validation_doubles.csv` is the full supplementary table: **every** within-10 pair, with `tier` blank when not selected (`coverage / validation / novel` otherwise), Costanzo DMF $\pm$ SD, derived SE, ε , p , the **Kuzmin 2018/2020 DMF $\pm$ SD** columns, and the strain id. 45 rows = $\binom{10}{2}$; kept as a CSV rather than inlined because its roughly 15 columns are too wide to render at this page width.

8 What was actually built

Of the 14 doubles this selection called for, **13 were constructed. YKL033W-A $\times$ YJR060W (CBF1) failed to construct** (reported by Aurosish Sharma on handover of the 2026-08-06 plating). It is absent from the run-4 strain list, so the built panel is 12 singles + 13 doubles + WT. The missing pair is a construction failure, not a dropout to impute.

The plating that resulted is recorded in `experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/PROVENANCE.md`: three independently randomized plates, WT at 28 wells and each mutant at 14 wells per plate, with both constituent singles of every double present on the same plate so $\varepsilon = f_{ab} - f_a f_b$ is computable within-plate.

9 Related notes

- `[[experiments.010-kuzmin-tmi.scripts.topk_triples_from_constructed_10]]`
- `[[experiments.010-kuzmin-tmi.scripts.optimized_doubles_setcover_constructed_10]]`
- `[[experiments.010-kuzmin-tmi.scripts.constructed_10_dmf_reference]]`
- `[[experiments.010-kuzmin-tmi.12_panel_crispr_fitness_assay]]` — the assay these doubles validate